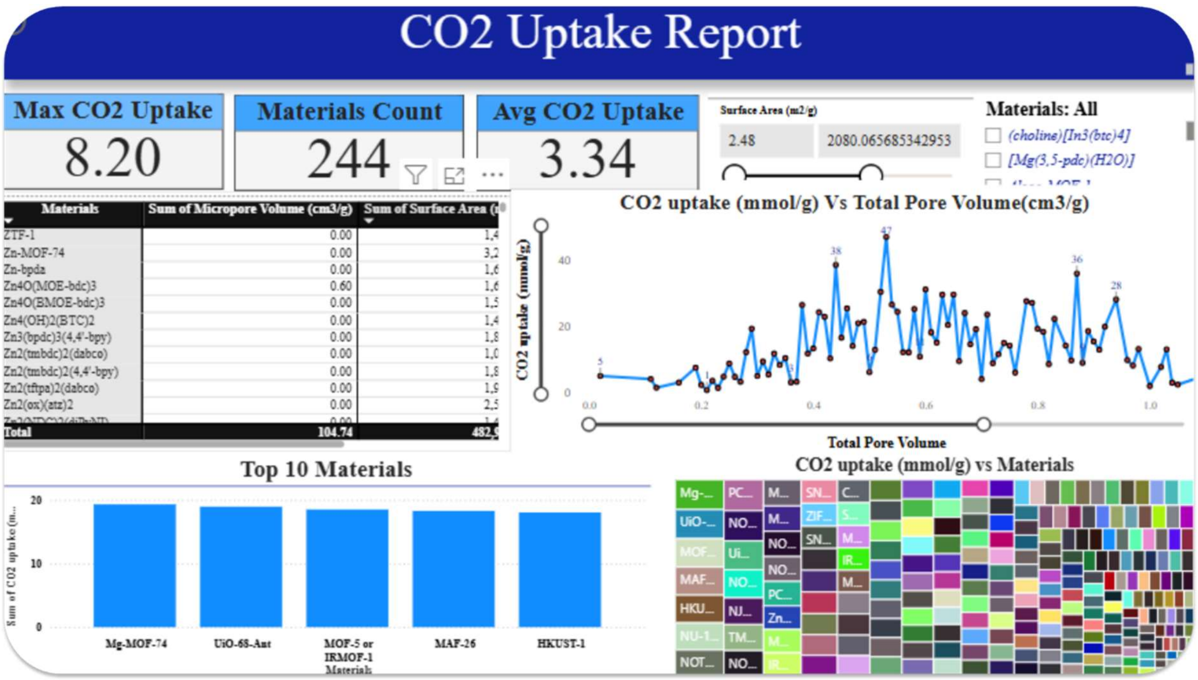
Machine Learning-Guided Design of Porous Carbons for CO_2 Capture: From Data Cleaning, ETL, preprocessing to Predictive Modeling



Research Group (GCRG), University of Calgary

Provided by: Amir Narimani

Date: Sept 2025

Abstract

Porous materials are widely recognized for their strong potential in carbon dioxide (CO_2) capture, making them promising candidates for large-scale industrial carbon mitigation. The adsorption efficiency of these materials is strongly influenced by factors such as temperature, pressure, pore size, and surface area. With the continuous rise in atmospheric CO_2, a major driver of climate change, developing optimized porous carbons for efficient CO_2 capture is increasingly important. Traditional experimental approaches for tuning textural and chemical properties are often time-consuming and resource-intensive. To overcome these limitations, this study employs machine learning (ML) to predict CO_2 adsorption capacity and guide the design of high-performance porous carbons. A comprehensive dataset, including pore volume, mean pore diameter, adsorption temperature and pressure, and BET surface area, was compiled. We applied a variety of data preprocessing techniques and trained four ML models, Gradient Boosting Regressor (GBR), Convolutional Neural Networks (CNN), Multi-Layer Perceptron (MLP), and Long Short-Term Memory (LSTM) networks using five-fold cross-validation. Feature engineering was performed to create additional predictive variables, improving model accuracy compared with the original dataset. The optimized models achieved an R2 score of 90.7% on the training set and 85.73% on the test set. Analysis revealed that the carbon-to-pressure ratio, temperature, and other material-specific physical properties are the most influential factors in CO_2 adsorption. These findings provide valuable insights for optimizing textural and adsorption characteristics, accelerating the discovery and design of high-performance adsorbents, and offering a roadmap for sustainable carbon capture solutions.

Introduction

Global warming and climate change have created major challenges for human society, including rising sea levels, accelerated ice melt, and an increased frequency of extreme weather events such as floods and droughts [1]. These issues are primarily driven by anthropogenic activities. Industrial operations, including oil and gas extraction, transportation, and power generation release significant amounts of greenhouse gases (GHGs) such as carbon dioxide (CO_2), methane (CH_4), and nitrogen oxides (NO_x) into the atmosphere [2]. Among these, CO_2 and CH_4 are considered the most influential contributors to global warming due to their strong greenhouse effect [3]. GHGs trap a fraction of the Earth's outgoing heat, preventing it from escaping into space. CO_2, in particular, absorbs radiation over a wide wavelength range (2000–15,000 nm), which overlaps with infrared energy. As CO_2 molecules absorb this energy, they vibrate and re-radiate it, with roughly half of the energy being emitted back toward Earth, intensifying the greenhouse effect [4]. One of the key strategies to mitigate CO_2 emissions is Carbon Capture and Storage (CCS). These technologies are designed to capture CO_2 emissions from major sources, such as power plants and industrial facilities, before they are released into the atmosphere, enabling their subsequent storage or utilization [6]. The development of high-performance adsorbents, particularly activated carbons, is essential for improving the efficiency and effectiveness of CO_2 capture processes [6,7]. CCS technologies are generally classified into three main categories: pre-combustion, oxy-fuel combustion, and post-combustion capture.

Carbons are highly porous materials derived from carbon-rich precursors, widely studied for their exceptional adsorption properties [11–13]. Their unique structure, characterized by a high surface area and a network of micro- and mesopores, makes them suitable for a range of applications, including CO_2 capture, environmental remediation, energy storage, catalysis, and air purification [25–28, 32]. Activated carbons can be synthesized through two main approaches: physical activation and chemical activation [14]. Physical activation involves a two-step process: carbonization followed by activation. During carbonization, the precursor is heated at high temperatures (typically 400–1000 °C) in an inert atmosphere (e.g., nitrogen or argon), which removes volatile components and converts the precursor into carbon-rich char. Chemical activation involves impregnating the precursor with chemical agents such as phosphoric acid (H_3PO_4), potassium hydroxide (KOH), or zinc chloride ($ZnCl_2$) prior to carbonization. This method allows lower activation temperatures and produces carbons with more developed pore structures [15–17]. Sustainable alternatives, like potassium carbonate (K_2CO_3), have also been explored to reduce the environmental impact of strong activators [18].

The performance of carbon porous materials in CO_2 capture is primarily determined by their surface area, pore volume, pore size distribution, and textural features [19]. Generally, a higher surface area provides more active sites for adsorption, while microspores enhance CO_2 uptake at low pressures. Mesoporous materials facilitate diffusion and can improve overall adsorption kinetics. Hybrid approaches combining functionalized solvents with mesoporous carbons have shown potential to reduce regeneration energy and enable thermal energy recovery, enhancing process efficiency [20].

Several key parameters influence CO_2 adsorption on activated carbons: Pressure, Higher pressures increase CO_2 uptake by providing a greater driving force for molecules to enter the pores. For example, adsorption at 20 bar is significantly higher than at lower pressures due to a shift from pore-filling mechanisms at low pressures to surface interactions at high pressures [20].

Temperature, CO_2 adsorption typically decreases with rising temperature (e.g., from 273 K to 298 K), as higher kinetic energy reduces the affinity of CO_2 molecules for the carbon surface [71]. Surface area and activation, The use of strong chemical activators like KOH or NaOH can dramatically increase BET surface areas (up to ~4000 m^2/g), enhancing CO_2 adsorption capacity [20].

carbons materials not only demonstrate high CO_2 adsorption but also good selectivity over other gases such as nitrogen (N_2) and methane (CH_4). This makes them highly effective for carbon capture in industrial settings. Their versatility extends to energy storage applications as electrode materials in supercapacitors and batteries, as well as catalyst supports for various chemical reactions [18]. Optimizing CO_2 adsorption using activated carbons is a complex and resource-intensive process. Achieving maximum capture efficiency requires careful tuning of material properties and operational conditions, such as surface area, pore structure, temperature, and pressure. Traditional experimental methods are often slow, costly, and labor-intensive, which limits the pace of innovation in carbon capture technologies.

Artificial intelligence (AI) and machine learning (ML) offer powerful solutions to these challenges. By leveraging data-driven modeling, AI can accelerate material discovery, predict adsorption performance, and optimize synthesis and operational parameters. For example, ML algorithms can analyze critical factors such as specific surface area, pore volume, and nitrogen functionalization to identify synthesis conditions that maximize CO_2 uptake [1-5]. Incorporation of nitrogen-containing functional groups enhances CO_2 adsorption via acid–base interactions, which can be efficiently predicted and optimized using ML techniques. Beyond material-level optimization, AI also enables real-time monitoring and system-level process optimization, improving operational efficiency and energy sustainability in biomass-based polygeneration plants with integrated carbon capture and utilization [17].

In recent years, the use of ML in CO_2 and pollutant adsorption research has gained considerable attention [19,20]. A variety of algorithms, including Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANN), have been applied for prediction and classification in carbon capture and storage (CCS) processes [21–22]. Among these, tree-based models such as Decision Trees (DT), Random Forests (RF), Gradient Boosting Decision Trees (GBDT), and Extreme Gradient Boosting (XGBoost) are particularly popular due to their robustness with smaller datasets and their ability to handle noisy data [23]. These models use recursive partitioning to minimize prediction errors and improve accuracy.

In this study, we implemented four ML models: Gradient Boosting Regressor (GBR), Convolutional Neural Networks (CNN), Multi-Layer Perceptron (MLP), and Long Short-Term Memory (LSTM) networks to predict CO_2 adsorption capacity of materials-derived porous carbons. After hyperparameter optimization, GBR and CNN achieved the highest predictive performance. The aim of this work is to develop predictive models that can guide the design of high-performance porous carbons, reducing the reliance on time-consuming experiments. Our analysis identified the carbon-to-pressure ratio and temperature as the most influential factors affecting CO_2 adsorption, followed by structural properties such as pore size distribution and surface area. These insights provide a data-driven roadmap for optimizing both material properties and operational conditions, ultimately enhancing CO_2 capture efficiency and supporting sustainable carbon mitigation strategies.

2. Methodology

In this study, experimental data for various carbon porous materials adsorbents were collected and used to develop machine learning (ML) models capable of predicting CO_2 adsorption capacity. The dataset included parameters such as pore volume, mean pore diameter, adsorption temperature, adsorption pressure, and BET surface area, with CO_2 uptake designated as the target variable. The main goal was to design algorithms that not only predict adsorption performance but also assess the influence of each parameter on CO_2 capture efficiency. Key considerations in adsorbent development included high adsorption potential, selectivity, operational stability, cost-effectiveness, reusability, ease of recovery, and rapid adsorption–desorption kinetics. Using three-dimensional plots generated by the most accurate ML model, the combined effects of these parameters on CO_2 uptake were analyzed and compared with previous work to evaluate the predictive capability of the developed models.

The methodology for model development comprised several key steps. First, all collected data were thoroughly reviewed to ensure objectivity, without introducing any assumptions regarding reliability. Next, the core features were categorized into two groups: morphological characteristics, which included specific surface area (BET, m^2/g), mean pore diameter (nm), and pore volume (cm^3/g); and adsorption conditions, such as temperature and pressure relevant to the CO_2 adsorption experiments. Finally, the target variable was defined as the CO_2 adsorption capacity of the carbon porous-based adsorbents under the various operational conditions considered in the study. This structured approach enabled the development of robust ML models that can accurately predict CO_2 uptake while providing insights into the relative importance of structural and operational parameters for optimized adsorbent design.

2. Data Preparation

Following data collection, preprocessing was performed to ensure the dataset was suitable for machine learning (ML) applications. Key steps included feature identification, normalization, and outlier detection.

2.1. Data Collection and Description

CO_2 adsorption data were compiled through an extensive literature review, focusing on carbon porous-based adsorbents. Textural properties (e.g., pore volume, pore size, and BET surface area) and CO_2 adsorption capacities were extracted from tables and figures in published studies. Adsorption capacities at pressures not explicitly reported were derived from adsorption isotherms using digitization software. A total of 527 data points were collected from 52 research articles, covering a range of operational conditions for carbon porous-based adsorbents. The dataset includes features such as pore volume, mean pore diameter, adsorption temperature, adsorption pressure, and BET surface area, with CO_2 uptake as the target variable.

Data Preprocessing and Feature Engineering

Preprocessing and feature selection are essential steps in data-driven methodologies, as they transform raw experimental data into a format suitable for machine learning analysis. Preprocessing cleans and standardizes the dataset, ensuring seamless integration into models and improving prediction accuracy. Feature selection further identifies the most relevant variables, reducing dimensionality and minimizing unnecessary computations.

To enhance predictive performance, additional features were engineered by combining or transforming existing variables. This approach allows the models to capture complex interactions between morphological properties and operational conditions, thereby improving the ability of machine learning algorithms to accurately predict CO_2 adsorption capacity. Input features were carefully identified and categorized into two main groups. Morphological properties included specific surface area (BET, m^2/g), mean pore diameter (nm), and pore volume (cm^3/g), while operational conditions encompassed adsorption temperature and pressure corresponding to the experimental setup. The target variable was defined as the CO_2 adsorption capacity of the carbon porous-based adsorbents under the corresponding conditions.

Feature Scaling

Feature scaling is an essential preprocessing step in many machine learning workflows. Although tree-based algorithms are generally robust to differences in feature scale, scaling can still offer advantages for other model types by smoothing the gradient descent process and facilitating faster convergence toward the loss function's minimum. In this study, we applied the Standard Scaler to the dataset to normalize feature distributions. The impact of this preprocessing step is illustrated in Fig. 3.

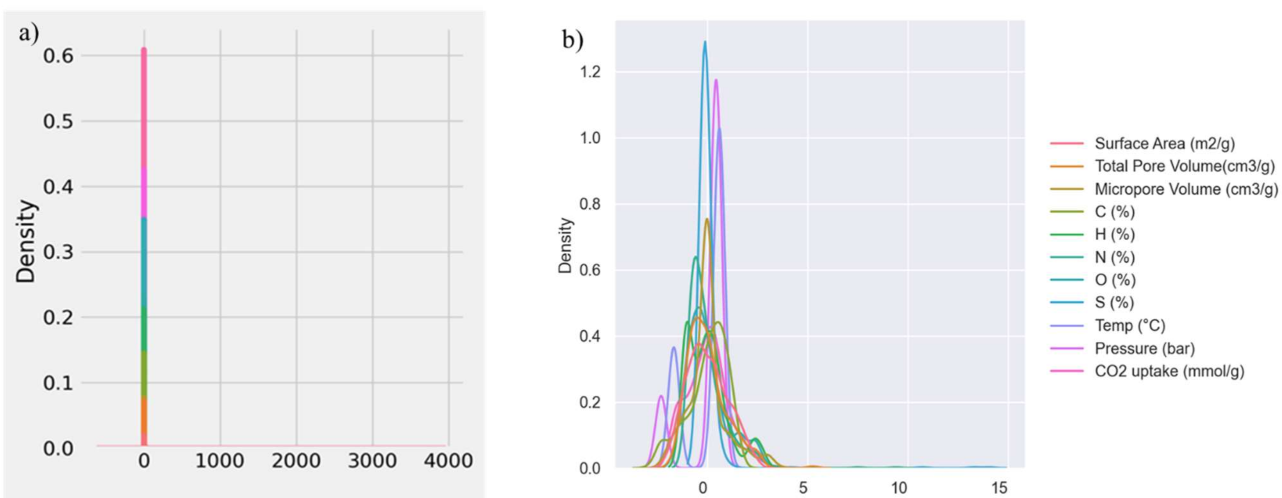


Fig. 3. The effect of applying the normal scaler on the dataset(a) without scaler effect and (b) with scaler effect.

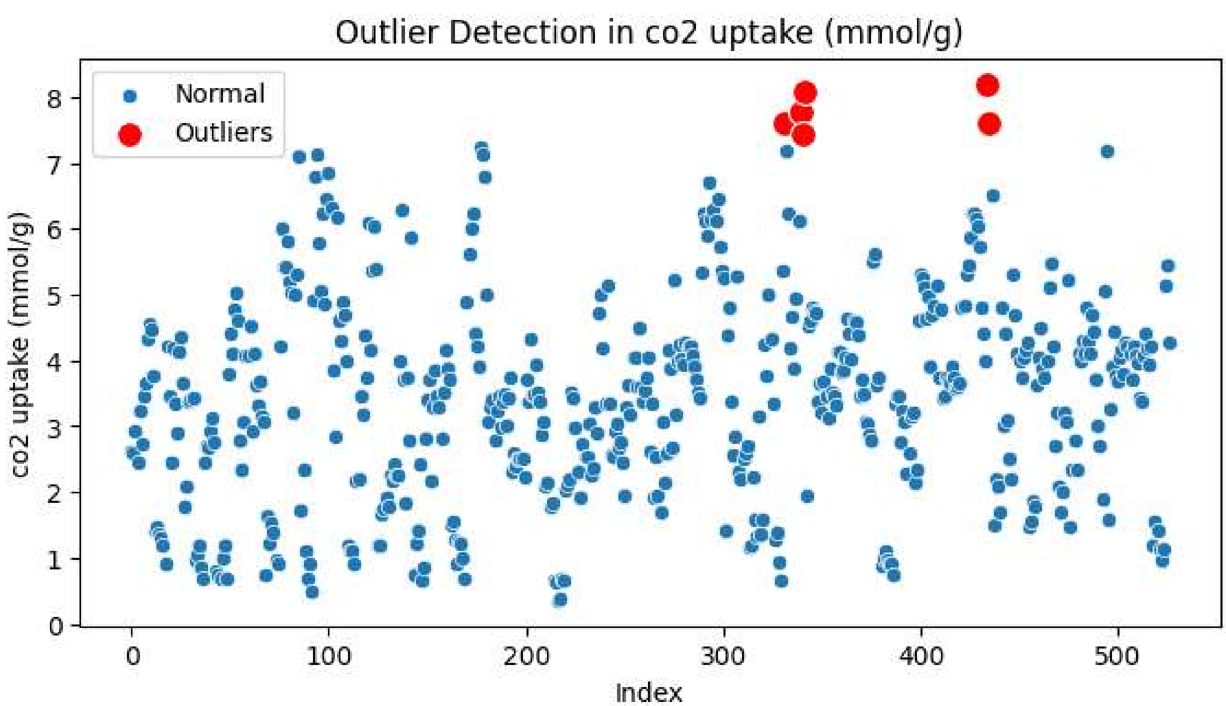
Imputing Missing Values

As shown in Table 2, the total pore volume (TPV) and microspore volume (MV) features contain 28 and 210 missing values, respectively. To address this issue, we initially applied the Newton

Interpolating Polynomial [17], which uses divided differences to construct a polynomial and is particularly suitable for handling non-uniformly spaced data points. However, our evaluation with advanced machine learning algorithms revealed that Newton interpolation produced suboptimal results, negatively impacting model accuracy. Consequently, we adopted Gradient Boosting Regressor (GBR)-based imputation, which significantly improved the predictive performance of all models.

2.2.3. Outlier Detection

Outliers can negatively impact model training and predictive accuracy. To identify anomalous data points, a distance-based approach was applied to the normalized feature space. First, all features were standardized using Z-score normalization. The center of the dataset, $\bar{\mu}$, was then computed as the mean vector of all standardized features. For each data point x_i, the Euclidean distance from the center was calculated as $d_i = \|x_i - \bar{\mu}\|_2$. A threshold for outlier detection was defined based on the 99th percentile of all distances, with any data points exceeding this threshold flagged as outliers. Using this procedure, four outliers were identified and removed, leaving a total of 454 data points for subsequent model training, validation, and testing. The outlier detection process is illustrated in Fig. 2.



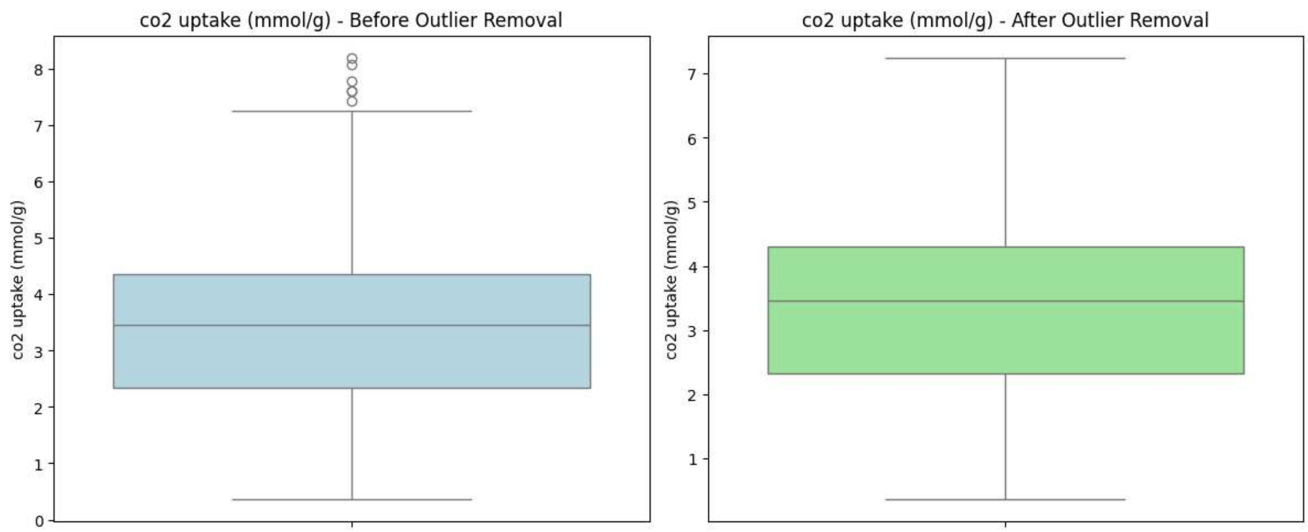


Fig.2. (a) Outlier detection based on Euclidean distance from center, (b) remove the outliers

Features correlation

The Pearson correlation coefficient between two features, x and y (r_{xy}), is defined as [22]. The coefficient takes a value between –1 and 1 and quantifies both the strength and direction of the linear relationship between the two variables. A value of –1 represents a perfect negative linear correlation, where an increase in x corresponds to a proportional decrease in y. A value of 0 indicates no linear correlation, while a value of 1 represents a perfect positive linear correlation, meaning that x and y increase together at a constant rate.

Statistical analysis was conducted on the slit pore carbon characteristics and CO_2 adsorption data, and Pearson correlation coefficient (PCC) analysis was performed between any two variables to calculate their quantitative linear correlation as follows:

$$\rho_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\sum_{i=1}^{n}(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \quad (7)$$

where ρ_{xy} is the PCC value from the feature to the target, $\bar{x}$ and $\bar{y}$ are the average values of the input characteristic x and the output target y, respectively. ρ_{xy} ranges from -1 to 1, where 0 indicates no linear correlation, and high positive or negative values indicate strong positive or negative correlation, respectively.

The correlation analysis (fig) reveals that pressure, surface area, and micropore volume are positively correlated with CO_2 adsorption across datasets. Interestingly, nitrogen content exhibits a dataset-dependent effect: it is positively correlated with CO_2 uptake in the dataset but negatively correlated in the dataset. In contrast, carbon content consistently shows a negative correlation with CO_2 adsorption across both datasets. Overall, these results suggest that pressure, surface area, micropore volume, nitrogen content, and carbon content are among the most influential features governing CO_2 capture performance.

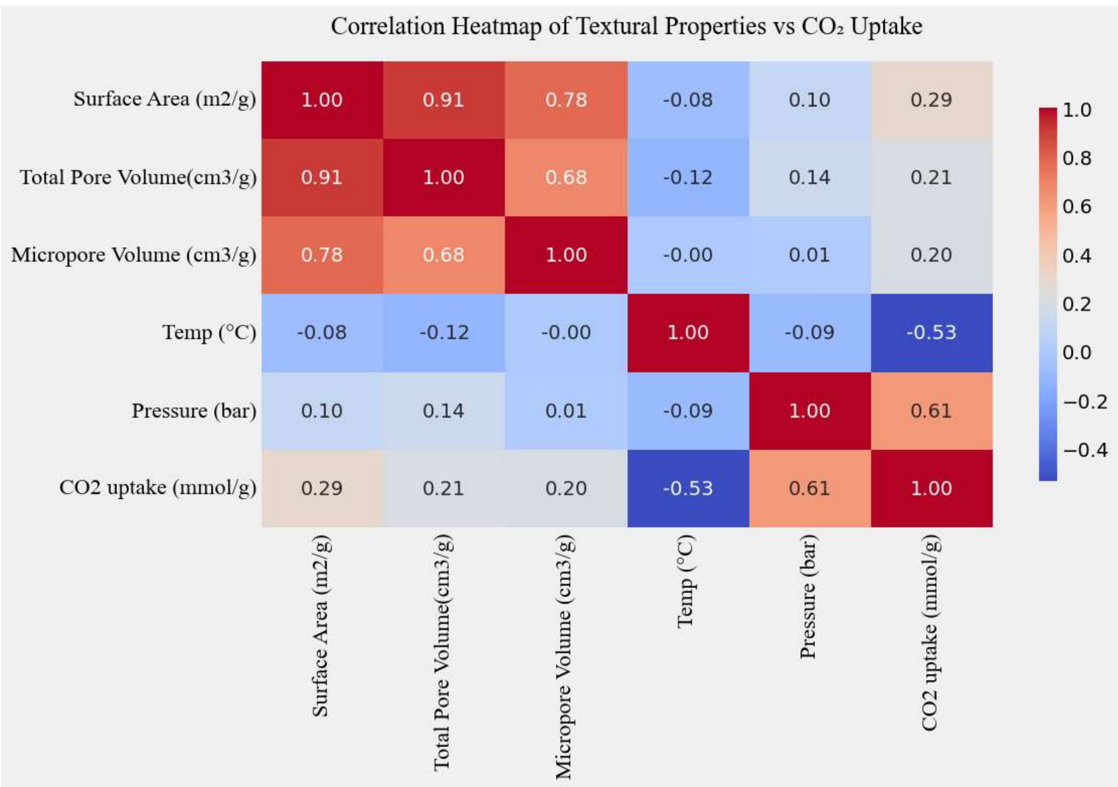


Fig. Pearson correlation of features

Multilayer Perceptron Artificial Neural Network (MLP-ANN)

A multilayer perceptron (MLP) is a type of artificial neural network composed of three types of layers: an input layer, one or more hidden layers, and an output layer (Fig. 3). The structure of the MLP, including the number of hidden layers and the number of neurons per layer, is determined using optimization algorithms tailored to the problem at hand. Each neuron in the network is connected to multiple neurons in adjacent layers, with weights assigned to represent the relative influence of each input. In the hidden layers, inputs are combined using weighted summation and then passed through an activation function, such as ReLU or sigmoid, to introduce non-linearity. The outputs from the hidden layers are subsequently forwarded to the output layer, where another transformation generates predictions for the target variable. Mathematically, the output of a single neuron can be expressed as:

$$y= f\left(\sum WiXi + b\right)$$

where y represents the neuron's output, f is the activation function, Wi are the weights assigned to each input Xi, and b is the bias term . In the context of CO_2 adsorption, MLP-ANNs are particularly valuable for capturing the complex, nonlinear relationships between sorbent properties (e.g., surface area, pore size, chemical composition) and adsorption performance metrics, such as CO_2 uptake capacity. During training, the network adjusts its weights using backpropagation combined with optimization algorithms like stochastic gradient descent or Adam to minimize prediction errors. Once trained, the MLP can generalize to predict the adsorption performance of new materials, reducing the need for extensive experimental testing and accelerating the design of high-performance adsorbents [15].

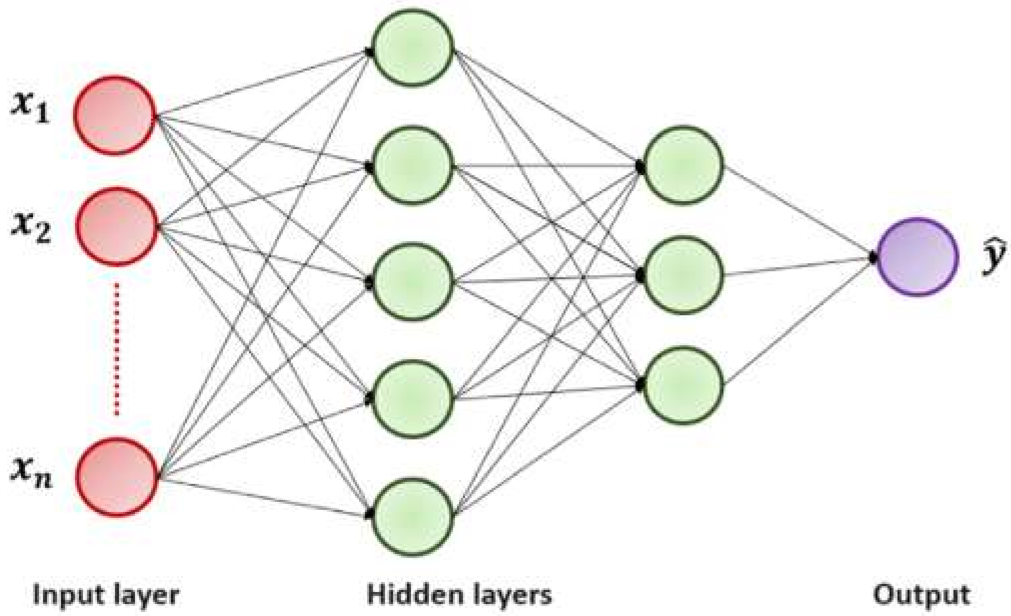


Fig. 3. The schematic view of the MLP-ANN.

Convolutional Neural Networks (CNN)

CNNs are composed of two main components: feature extraction layers and output layers, which provide either classification or regression results. The feature extraction component consists of convolution and pooling layers, where the convolution layers apply learnable filters to the input data to identify patterns relevant to the prediction task. Pooling layers are interspersed between successive convolution layers to downsample feature maps, reducing dimensionality, mitigating overfitting, and improving computational efficiency. Unlike conventional artificial neural networks, the neurons in convolution and pooling layers are not fully connected; each neuron only connects to a local region of the previous layer, resulting in fewer free parameters and lower computational demand. In regression applications, CNNs produce continuous output values. The input is typically represented as a feature matrix, with rows corresponding to individual data points and columns to features. The extracted features are then passed through one or more fully connected layers, which apply linear transformations followed by non-linear activations. The final output layer generates a continuous prediction for the target variable. During training, the network minimizes the mean squared error (MSE) between predicted and actual values, with weights and biases updated iteratively through backpropagation and gradient descent [43,44]. To optimize predictive performance, CNN architectures were systematically evaluated with varying numbers of layers, neuron counts per layer, and activation functions (Fig. 14). The Adam optimizer was used to update model parameters during training, leveraging adaptive learning rates to improve convergence. In our implementation, the CNN models consist of three hidden layers, each with the same number of neurons as the input features. The Rectified Linear Unit (ReLU) activation function is applied to all hidden layers to introduce non-linearity and accelerate training. The MSE loss function serves as the optimization objective, measuring the average squared difference between predicted and actual CO_2 adsorption values [23]. In summary, the CNN architecture effectively captures complex, nonlinear relationships between morphological and operational features and CO_2 uptake, providing robust and accurate predictions while remaining computationally efficient.

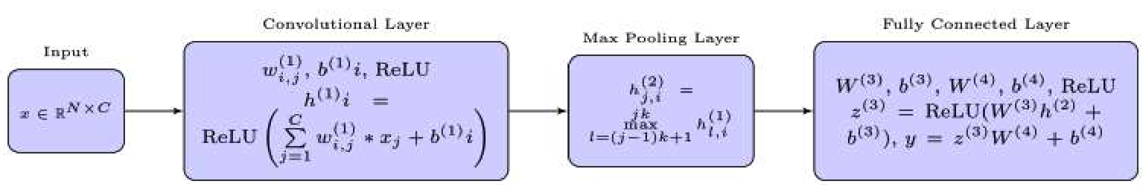


Fig. 14. 1D 3-layer CNN for regression.

Recurrent Neural Network: Long Short-Term Memory (LSTM)

Unlike traditional neural networks, where neurons in the same hidden layer are not interconnected, recurrent neural networks (RNNs) allow nodes within a hidden layer to communicate with each other. This means that the output at a given time step depends not only on the current input but also on the previous hidden state, enabling RNNs to capture sequential dependencies in data

$$\begin{aligned}
&\text{Input gate}: i_t = \sigma\left(W_{xi}X_t + W_{hi}h_{t-1} + b_i + W_{ci} \odot c_{t-1}\right) \\
&\text{Forget gate}: f_t = \sigma\left(W_{xf}X_t + W_{hf}h_{t-1} + b_f + W_{cf} \odot c_{t-1}\right) \\
&y_t = \tanh\left(W_{xy}X_t + W_{hy}h_{t-1} + b_y\right) \\
&c_t = f_t \odot c_{t-1} + i_t \odot y_t \\
&\text{Output gate}: o_t = \sigma\left(W_{xo}X_t + W_{ho}h_{t-1} + b_o + W_{co} \odot c_{t-1}\right) \\
&h_t = o_t \odot \tanh(c_t)
\end{aligned}$$

Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber (1997), enhance standard RNNs by incorporating memory units with adaptive gating mechanisms. These gates, input, forget, and output, control the extent to which information is stored, updated, or discarded in each memory cell, allowing the network to retain long-term dependencies while selectively forgetting irrelevant information. This mechanism mitigates the vanishing and exploding gradient problems commonly encountered in RNNs. For CO_2 adsorption prediction, LSTMs can be used to model temporal or sequential patterns in datasets where the adsorption capacity may depend on sequences of experimental conditions, material properties, or operating parameters. Let denote the predicted CO_2 adsorption sequence. The LSTM operations are defined as follows:

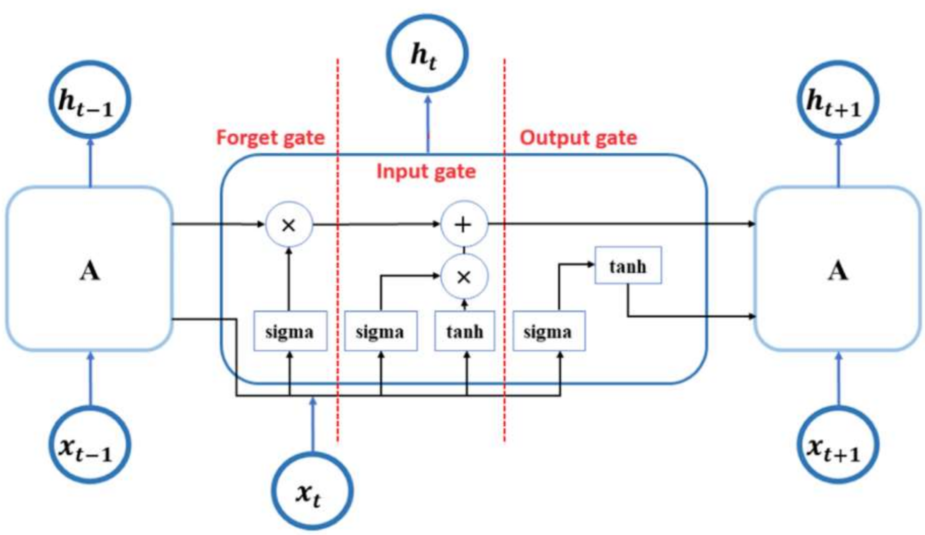


Fig. 4. Schematic view of an LSTM with three gates.

Here, σ is the logistic sigmoid function, ⊙ denotes elementwise multiplication, c_t represents the current cell state, h_t is the hidden state, b_* are bias vectors, and W_* are weight matrices updated during training via backpropagation through time. In the context of CO_2 adsorption, LSTMs allow the model to account for dependencies between material features, adsorption conditions, and sequential experimental variations, enabling improved prediction of adsorption capacity across different operating conditions.

Models Precision Determination

To assess the predictive performance of the machine learning models, several standard regression metrics were employed. These metrics provide quantitative measures of how closely the model predictions match the actual observed values.

Coefficient of Determination (R2)

The metric indicates the proportion of variance in the target variable that is explained by the model. Values of closer to 1 signify better model fit and higher predictive accuracy. It is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (Y_i^{actual} - Y_i^{pred})^2}{\sum_{i=1}^{N} (Y_i^{actual} - \overline{Y}_{ave}^{actual})^2} \qquad (8)$$

Root Mean Squared Error (RMSE)

RMSE is the square root of the MSE and provides a measure of the standard deviation of prediction errors. It is expressed in the same units as the target variable and gives an interpretable measure of prediction accuracy

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (Y_i^{actual} - Y_i^{pred})^2} \qquad (9)$$

Where Y_i^{actual} and Y_i^{pred} are the actual and predicted values, respectively, and $\overline{Y}_{ave}^{actual}$ is the average of the actual values.

2.6. Training and Validation

To ensure reliable evaluation of model performance and robust hyperparameter selection, a five-fold cross-validation (K-Fold) strategy was implemented. Initially, the dataset was split into a training set containing 80% of the data and a separate validation set comprising the remaining 20%, which was withheld from model training and hyperparameter optimization. Within the training set, data were further divided into five equally sized folds. During each iteration, four folds were used to train the model, while the remaining fold served as the evaluation set. This procedure was repeated five times so that each fold acted as the test set once. The average performance across all five iterations was used as the criterion for hyperparameter tuning. This approach mitigates bias from a single train-test split and improves generalization by exposing the model to multiple training and testing scenarios. Hyperparameter optimization is crucial for machine learning models such as GBR, ANN, CNN, and LSTM. Hyperparameters such as the number of hidden layers and neurons, activation functions, learning rates, batch size, dropout rates, and sequence lengths (for LSTM) are set prior to training and significantly influence model convergence, accuracy, and robustness. Exhaustive methods like grid search or stochastic approaches such as random search are commonly used to explore the hyperparameter space. For deep learning models, adaptive optimization algorithms like Adam or RMS prop were employed to update model weights while minimizing the loss function efficiently. This structured approach ensures that each model is trained under optimal conditions, balancing predictive accuracy with generalization capability, and enabling the development of reliable models for CO_2 adsorption prediction.

Porous Material Synthesis via Treatment and Autoclave

Porous materials can be synthesized using controlled chemical and thermal treatments combined with autoclave-assisted processes to tailor their structural and textural properties. The general procedure involves the following steps:

Precursor Preparation: A suitable precursor material (e.g., polymers, biomass, metal-organic frameworks, or inorganic compounds) is selected and pretreated to remove impurities or enhance reactivity.

Chemical Treatment: The precursor is exposed to chemical reagents such as acids, bases, or salts to modify surface chemistry, generate pores, or induce crosslinking. The treatment conditions, including concentration, temperature, and duration, are optimized to achieve the desired porosity and surface functionality.

Autoclave Processing: The chemically treated precursor is placed in an autoclave and subjected to controlled temperature and pressure conditions, often under inert or reactive atmospheres. This hydrothermal or solvothermal treatment facilitates pore formation, crystallization, and structural ordering, producing a stable porous network.

Post-Treatment and Activation: After autoclave processing, the material is washed to remove residual chemicals and dried under controlled conditions. Additional activation steps, thermal or chemical, may be applied to enhance surface area and pore connectivity further.

This combined approach allows precise control over pore size, volume, morphology, and surface chemistry, making the resulting porous materials suitable for a wide range of applications, including gas adsorption, catalysis, energy storage, and separation processes

Experiment for CO_2 Uptake

The CO_2 capture capacity of the porous material was measured using thermogravimetric analysis (TGA) and autoclave-assisted testing under controlled conditions. The procedure is as follows:

1. Pre-Treatment: The porous material was first vacuum-dried at 200 °C for 2 hours to remove moisture and impurities that could interfere with adsorption measurements.
2. Autoclave/Fixed-Bed Setup: For adsorption tests, the material (e.g., 1–2 g sample) was placed in a fixed-bed column or autoclave reactor. The system was purged with an inert gas (N_2) at 100 mL/min while heating to 400 °C for 2 hours to eliminate any previously adsorbed gases and stabilize the sample.
3. Adsorption Measurement: After cooling to room temperature (25–30 °C), a gas mixture containing 0.5% CO_2 in N_2 was introduced at a flow rate of 100 mL/min. The adsorption process continued until saturation was reached. CO_2 uptake was continuously monitored using either a non-destructive infrared CO_2 sensor (e.g., MH-Z19B) or the weight change recorded via TGA, providing real-time adsorption data.
4. Data Calculation:

- CO_2 Uptake (mg/g): Calculated from the weight gain of the sample in TGA or the breakthrough curve in the fixed-bed setup using the mass balance of the system.
- Breakthrough Analysis: Time-dependent CO_2 concentration data from the sensor were used to determine adsorption capacity and kinetics.

This approach allows accurate quantification of CO_2 adsorption, enabling comparison between different porous materials or experimental conditions, and provides essential data for modeling adsorption performance and optimizing material properties.

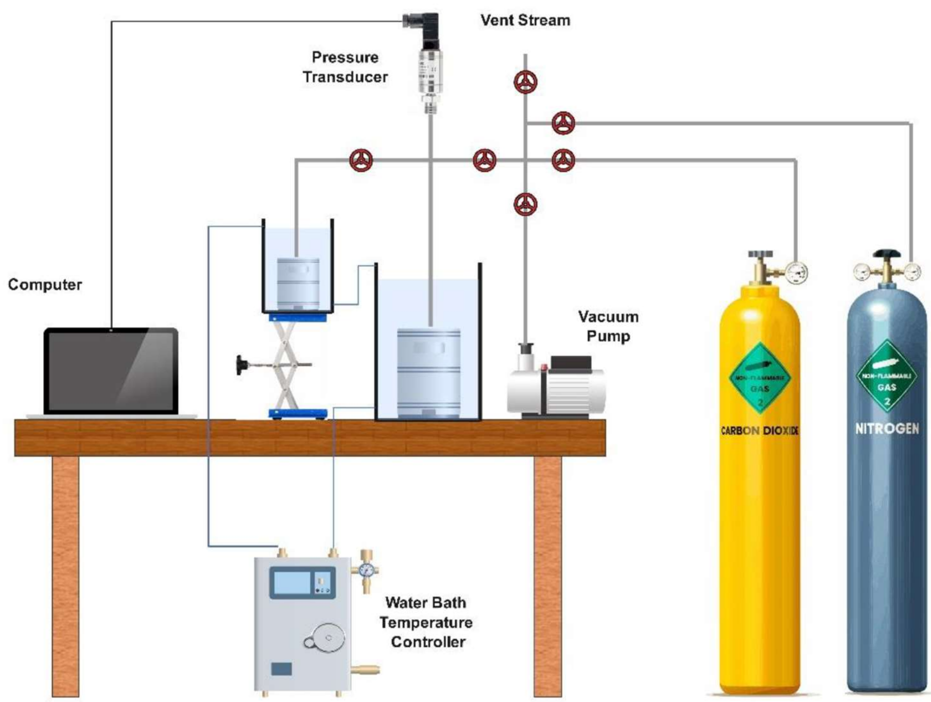


Fig.5. Schematic of experimental apparatus used in this study for CO2 capture investigation

Figure 3 illustrates the structural properties of carbon porous -based adsorbents along with their corresponding CO_2 adsorption capacities under different pressures and temperatures. The statistical distribution plots, generated after outlier removal, reveal that surface area, pore volume, and CO_2 uptake follow approximately unimodal distributions, with adsorption capacity exhibiting a slight right skew, indicating the presence of high-performing adsorbents. In contrast, mean pore diameter, pressure, and temperature distributions are right-skewed and appear discrete, reflecting the typical ranges employed in adsorption experiments. These patterns indicate that the dataset is broad and representative, providing a solid foundation for training machine learning models to accurately predict CO_2 adsorption across a variety of structural and operational conditions.

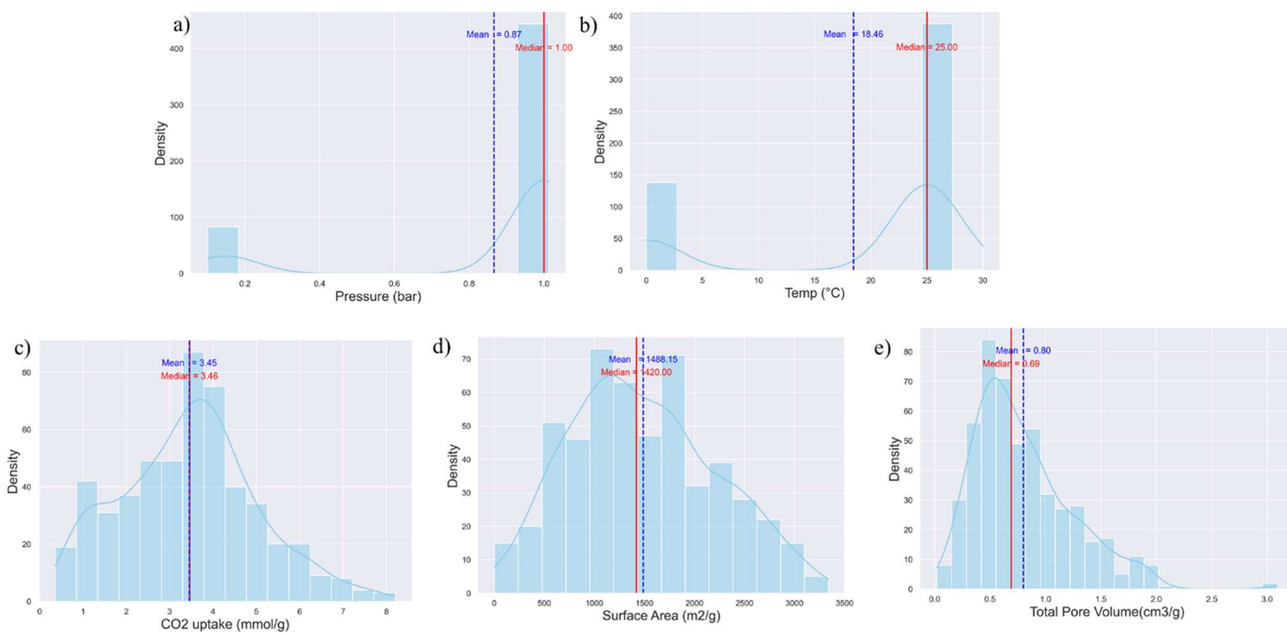


Figure 1. The statistical distribution of features: (a) Pressure, (b) Temp (∘C), (c) CO2 Uptake, (d) Surface Area (m2 /g), (h) Total Pore Volume (cm3/g).

Hyperparameters and cross-validation

In gradient boosting, the process iteratively adds new decision trees, with each tree trained to predict the residual errors of the current ensemble as the baseline. The final model prediction is obtained by summing the outputs of all trees. During training, the algorithm minimizes a specified loss function by adjusting the parameters of each newly added tree. The key hyperparameters for tree-based models include the number of estimators, maximum depth, learning rate, and choice of loss function. By default, the number of estimators (trees) is set to 100; increasing this number typically improves model performance but may lead to longer training times and an increased risk of overfitting. The maximum depth, commonly set to 3, controls tree complexity by limiting the total number of splits. The learning rate, with a default value of 0.1, plays a crucial role in gradient boosting by determining the step size during each iteration and guiding the model toward convergence. In this work, these hyperparameters were carefully tuned to develop a modified version of the Gradient Boosting Regressor (GBR), which demonstrated improved performance over the default configuration. After normalizing the dataset, the subsequent step is to implement a k-fold cross-validation (CV) approach to partition the data into training and testing subsets. This procedure involves running multiple training cycles, where the model is refined until it achieves the minimum possible error rate. Such repetition is essential for assessing both the stability and generalization performance of the machine learning model [22]. In k-fold CV, the dataset is divided into k portions in this case, five with each portion serving once as the test set while the others are used for training in successive rounds. This strategy plays a key role in addressing bias–variance trade-offs commonly encountered in ML. The choice of k (commonly 5 or 10) is influenced by the dataset size [44]. In this study, we selected $k = 5$. Figure 3 depicts the k-fold cross-validation procedure.

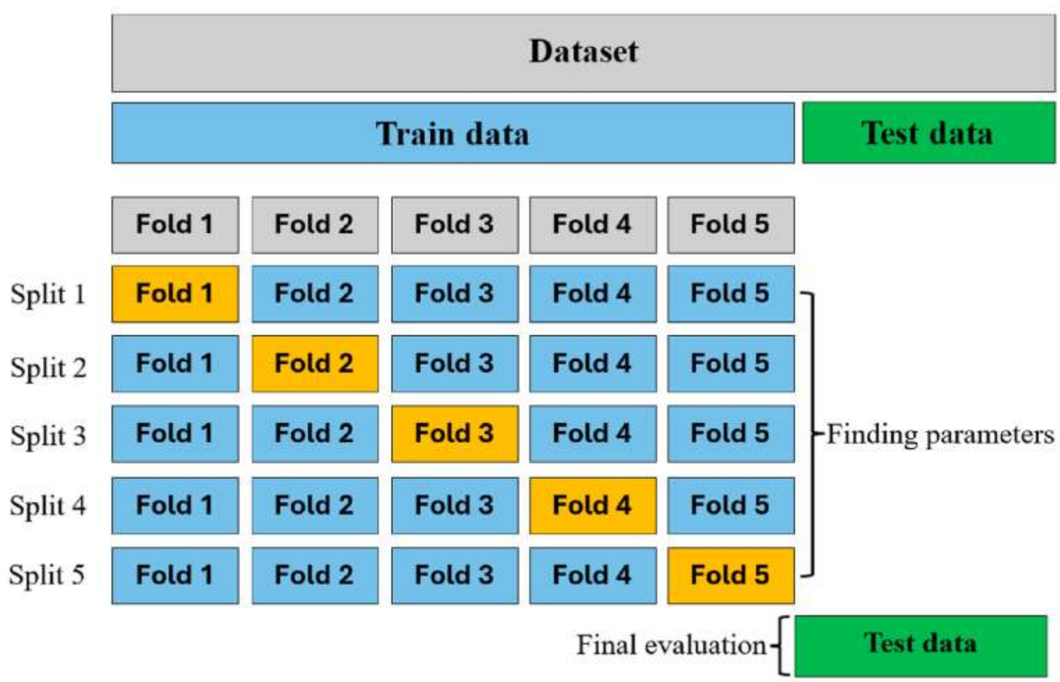


Fig. 3. The 5-fold CV technique.

The statistical distribution

As shown in Table 1, the regression model explains 68.0% of the variation in CO_2 uptake (R^2 = 0.680, Adj. R^2 = 0.674), and the overall F-test is highly significant ($F = 109.8$, $p < 0.0001$), confirming that the predictors jointly explain a substantial portion of the response variance. Among the predictors, pressure shows the strongest positive effect ($\beta = 2.856$, $t = 21.70$, $p < 0.0001$), indicating that an increase in pressure directly enhances CO_2 uptake, which is consistent with adsorption isotherm behavior. Temperature exerts a significant negative effect ($\beta = –0.070$, $t = –18.75$, $p < 0.0001$), showing that higher temperatures reduce uptake, in agreement with the exothermic nature of physical adsorption. Textural properties are also important. Surface area significantly increases CO_2 uptake ($\beta = 0.0008$, $t = 7.53$, $p < 0.0001$), while micropore volume has a positive effect as well ($\beta = 0.585$, $t = 2.52$, $p = 0.012$), confirming the critical role of micropores in CO_2 capture. In contrast, total pore volume has a significant negative coefficient ($\beta = –1.067$, $t = –6.07$, $p < 0.0001$), suggesting that larger pores do not contribute to adsorption and may reduce efficiency by lowering micropore fraction. Elemental composition (C, H, N, O, and S contents) does not show statistically significant effects (all $p > 0.19$), indicating that, within this dataset, chemical composition plays a negligible role compared to physical textural parameters. However, the skewness and kurtosis values reveal that nitrogen (Skew = 3.21, Kurtosis = 20.46) and sulfur (Skew = 12.27, Kurtosis = 157.85) contain extreme outliers, which likely explains their nonsignificance. Diagnostic statistics show that the model residuals deviate slightly from normality (Jarque–Bera test: $\chi^2 = 33.56$, $p < 0.0001$), and the high condition number (Cond. No. = 8.62×10^4) indicates potential multicollinearity among predictors such as surface area and pore volumes. Despite this, the Durbin–Watson statistic (DW = 1.04) suggests some positive autocorrelation in residuals, though this is less critical in cross-sectional adsorption data. In summary, the statistical evidence demonstrates that pressure, temperature, surface area, and micropore volume are the most significant determinants of CO_2 uptake, while total pore volume is detrimental and elemental composition is not predictive in this model.

Table1.The statistical distribution

VARIABLE	COEFFICIENT (B)	STD. ERROR	T-VALUE	P-VALUE	95% CI (LOWER)	95% CI (UPPER)
INTERCEPT (CONST)	3.9498	2.026	1.949	0.052	-0.031	7.931
SURFACE AREA (M^2/G)	0.0008	0.000	7.528	0.000	0.001	0.001
TOTAL PORE VOLUME (CM^3/G)	-1.0666	0.176	-6.073	0.000	-1.412	-0.722
MICROPORE VOLUME (CM^3/G)	0.5847	0.232	2.515	0.012	0.128	1.041
C (%)	-0.0238	0.020	-1.170	0.242	-0.064	0.016
H (%)	-0.0463	0.039	-1.194	0.233	-0.122	0.030
N (%)	-0.0020	0.023	-0.087	0.931	-0.048	0.044
O (%)	-0.0256	0.020	-1.292	0.197	-0.065	0.013
S (%)	-0.0344	0.046	-0.742	0.459	-0.125	0.057
TEMP (°C)	-0.0701	0.004	-18.747	0.000	-0.077	-0.063
PRESSURE (BAR)	2.8560	0.132	21.698	0.000	2.597	3.115

Deep Learning Models

As illustrated in Fig. 13, the CNN model consistently outperforms the other deep learning approaches. Consequently, CNN was chosen as the primary algorithm for final predictions. Technical readers are referred to [42] for detailed explanations of MLP and LSTM architectures. Each deep learning model and its corresponding pipeline were executed seven times to account for variability in training outcomes caused by factors such as weight initialization and differences in optimization pathways. Each run produced slightly different R^2 scores, reflecting the stochastic nature of neural network training. Fig. 13 compares R^2 values achieved by MLP1 (2 hidden layers), MLP2 (3 hidden layers), LSTM, and CNN, highlighting CNN's superior predictive capability. While careful preprocessing can enhance model performance, Fig. 12 indicates a potential risk of overfitting, evidenced by the gap between training and validation loss curves. Although LSTM is primarily suited for sequential data, its performance here was lower than both MLP and CNN models, confirming its limited applicability for this dataset. The CNN model's advantage stems from its ability to extract localized and hierarchical features from structured datasets, which may not be immediately apparent from raw input variables. Convolutional layers effectively identify the most relevant features for the regression task, enhancing predictive accuracy [46]. Fig. 15 presents CNN predictions across four datasets, with Fig. 16 showing the corresponding training and validation loss curves. Comparison with Fig. 12 confirms that CNN achieves higher predictive accuracy and improved generalization. Fig. 17 further demonstrates that CNN predictions closely match actual measurements, indicating strong reliability and robustness for CO_2 adsorption prediction. The model evaluation included analyses of the loss function and Mean Absolute Error (MAE) for both training and testing datasets, along with prediction results on the test set, demonstrating CNN's superior performance and consistency across multiple experimental runs.

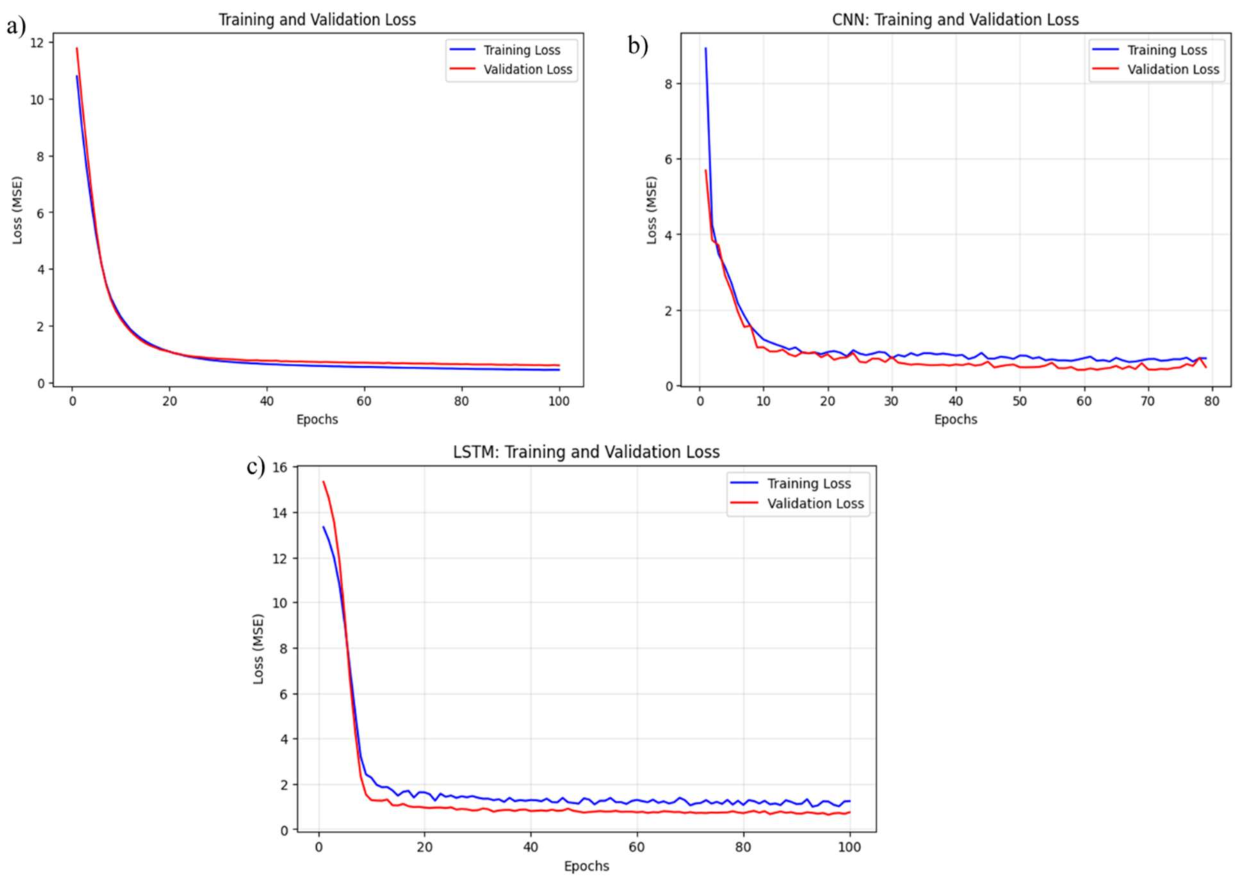


Fig. 16. The loss function for both training and testing datasets using a) ANN, b) CNN and c) LSTM algorithm .

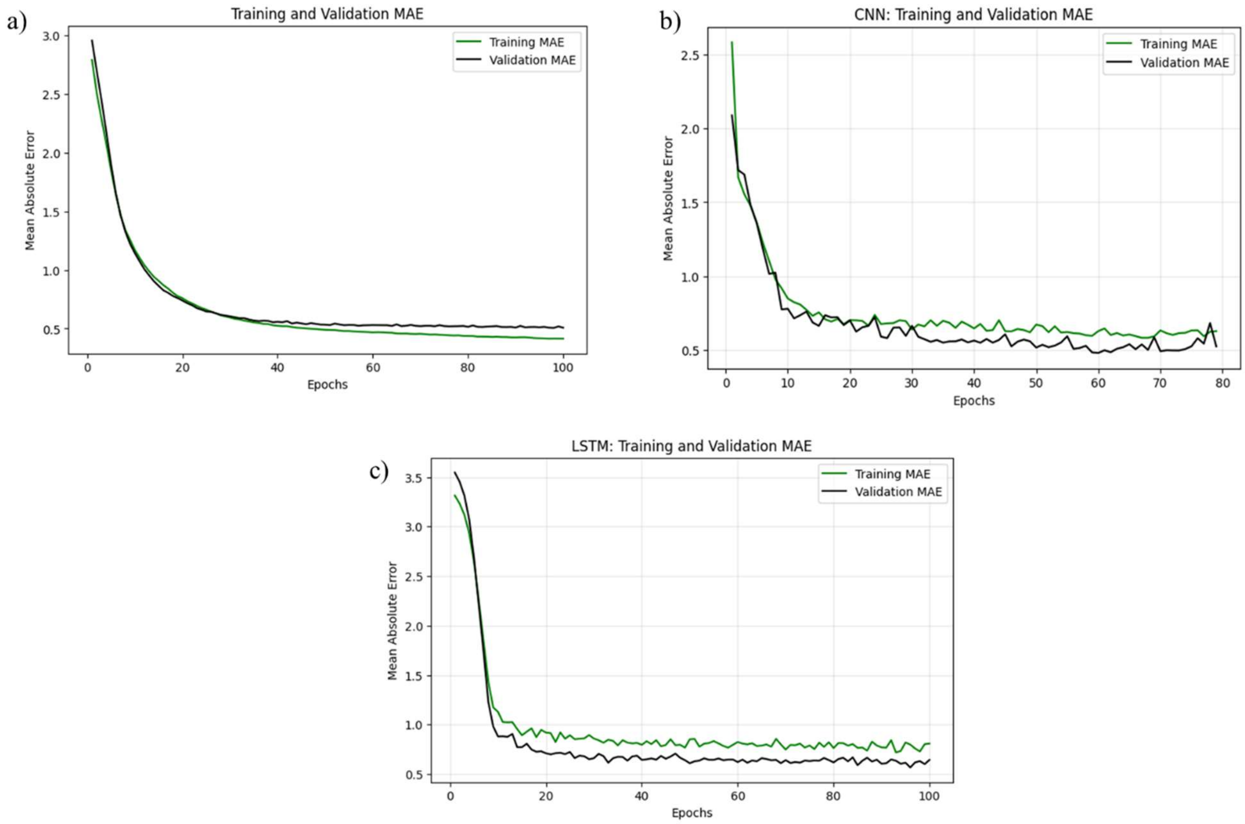


Fig. 16. The MAE function for both training and testing datasets using a) ANN, b) CNN and c) LSTM algorithm.

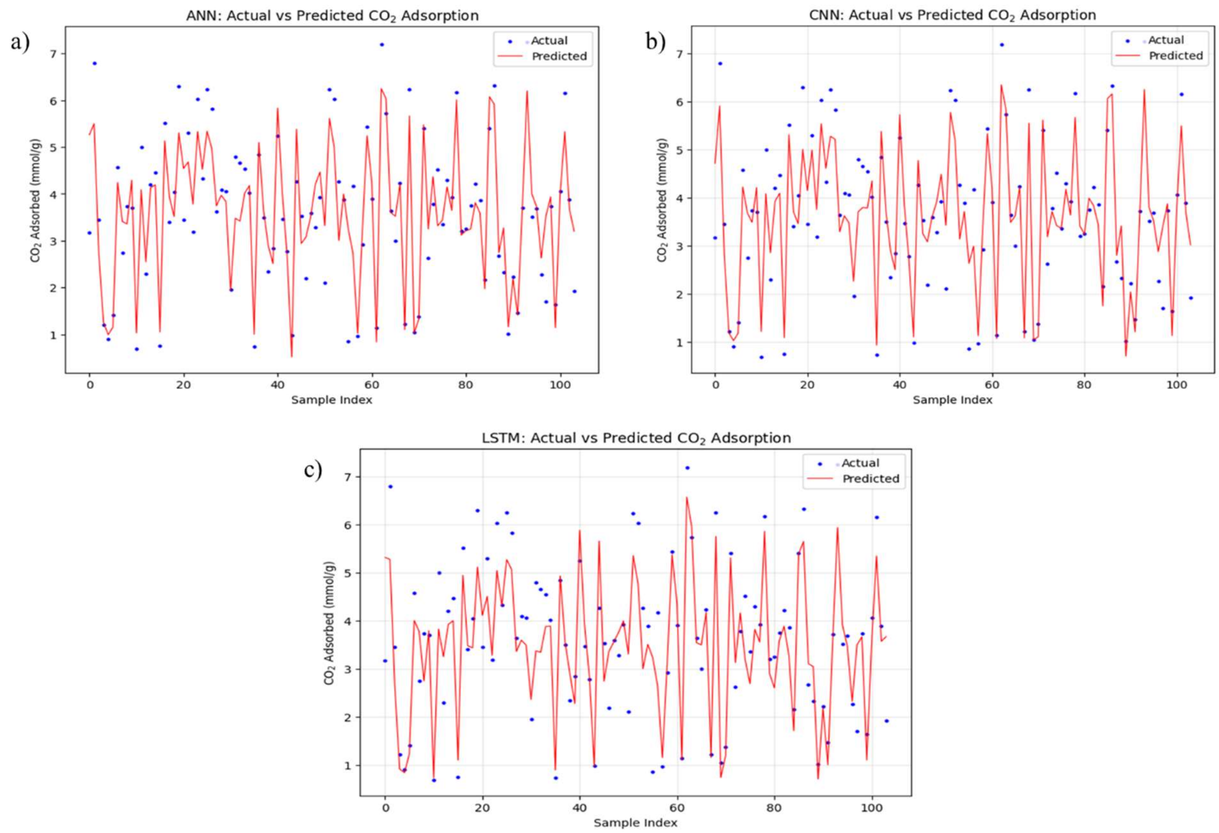


Fig. 17. The prediction results on the test dataset using a) ANN, b) CNN and c) LSTM algorithm.

The predictive accuracy of each model was visually assessed through actual versus predicted uptake plots, where data points closest to the y = x reference line indicate the highest accuracy. In addition, residual histogram plots were generated for both training and validation sets to evaluate the distribution of prediction errors. A kernel density estimate (KDE) curve (Figure) was overlaid on each histogram to provide a smoothed view of the residual distribution. Well-performing models exhibit residuals that are symmetrically distributed around zero, with minimal skewness and few outliers, indicating accurate predictions and strong generalization. These graphical and statistical analyses were used to systematically compare model performance and identify the most reliable approach for CO_2 adsorption prediction. The predictive performance of machine learning models for CO_2 adsorption capacity (Y) was assessed using both training and validation datasets, with evaluation metrics including Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), as shown in Figure . Model predictions were compared to actual values using the ideal parity line (Y = X) to visually assess accuracy and potential overfitting. The Artificial

Neural Network (ANN) effectively captured the dominant patterns in the training data, achieving MSE = 0.3839, RMSE = 0.6196, MAE = 0.4541, and R^2 = 0.9292. However, in the validation phase, performance declined (MSE = 0.7718, RMSE = 0.8785, MAE = 0.6551, R^2 = 0.8919), with predictions slightly underestimating higher Y values, indicating moderate overfitting. Residuals were approximately centered around zero but displayed slight right skewness, reflecting sensitivity to unseen data variability. The Long Short-Term Memory (LSTM) network captured sequential dependencies in the features, with training metrics of RMSE = 0.5910, R^2 = 0.9380. Validation performance dropped slightly (RMSE = 0.8298, R^2 = 0.8995), and residuals showed moderate spread at mid-to-high Y values, suggesting some sensitivity to data variability and a slightly higher risk of overfitting compared to tree-based methods. The Convolutional Neural Network (CNN) achieved high predictive accuracy, with training metrics of MSE = 0.2967, RMSE = 0.5448, MAE = 0.4123, R^2 = 0.9482 and strong validation performance (MSE = 0.5723, RMSE = 0.7565, MAE = 0.5710, R^2 = 0.9175). Residuals were tightly clustered around zero with minimal skew, indicating effective modeling of complex nonlinear relationships without significant overfitting. Overall, CNN, outperformed ANN and LSTM in both training and validation. Among them, CNN-LSTM offered the optimal balance of predictive accuracy, generalization, and robustness, effectively capturing complex interactions between structural and operational factors influencing CO_2 adsorption while minimizing overfitting.

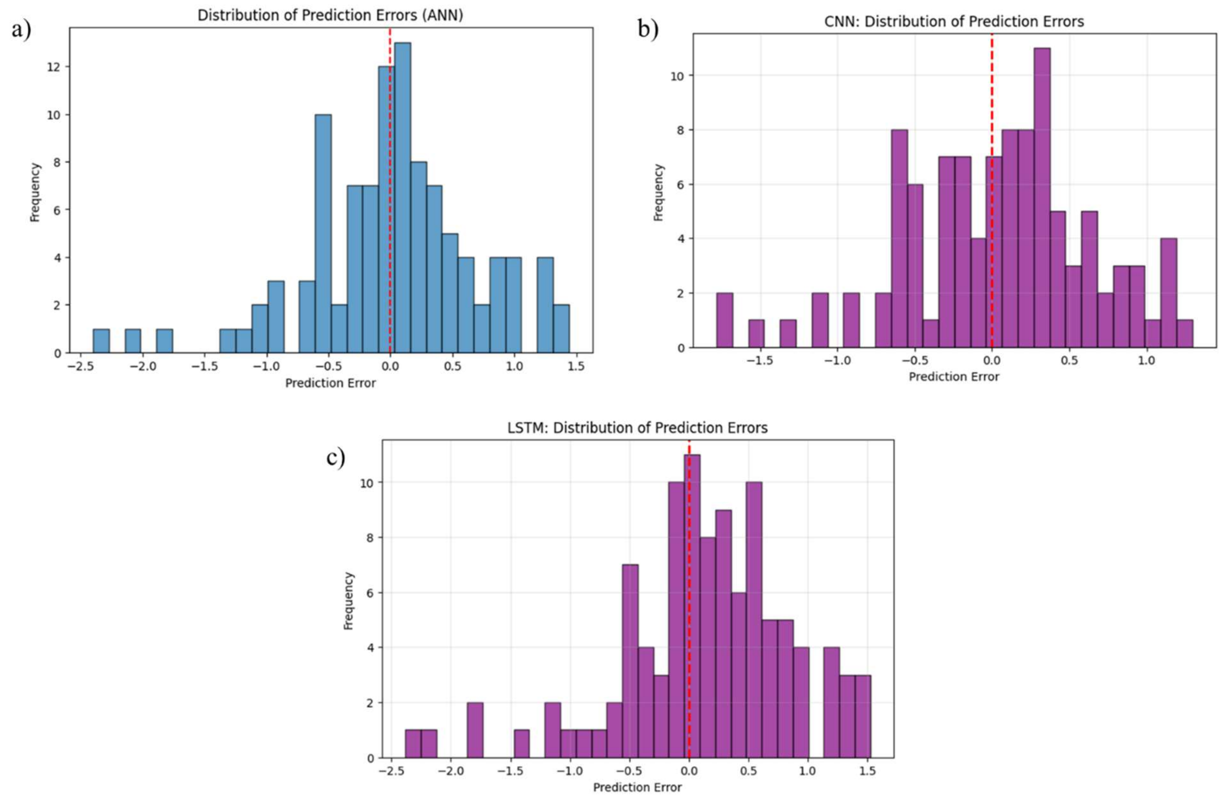


Figure. Kernel density estimate (KDE) curve

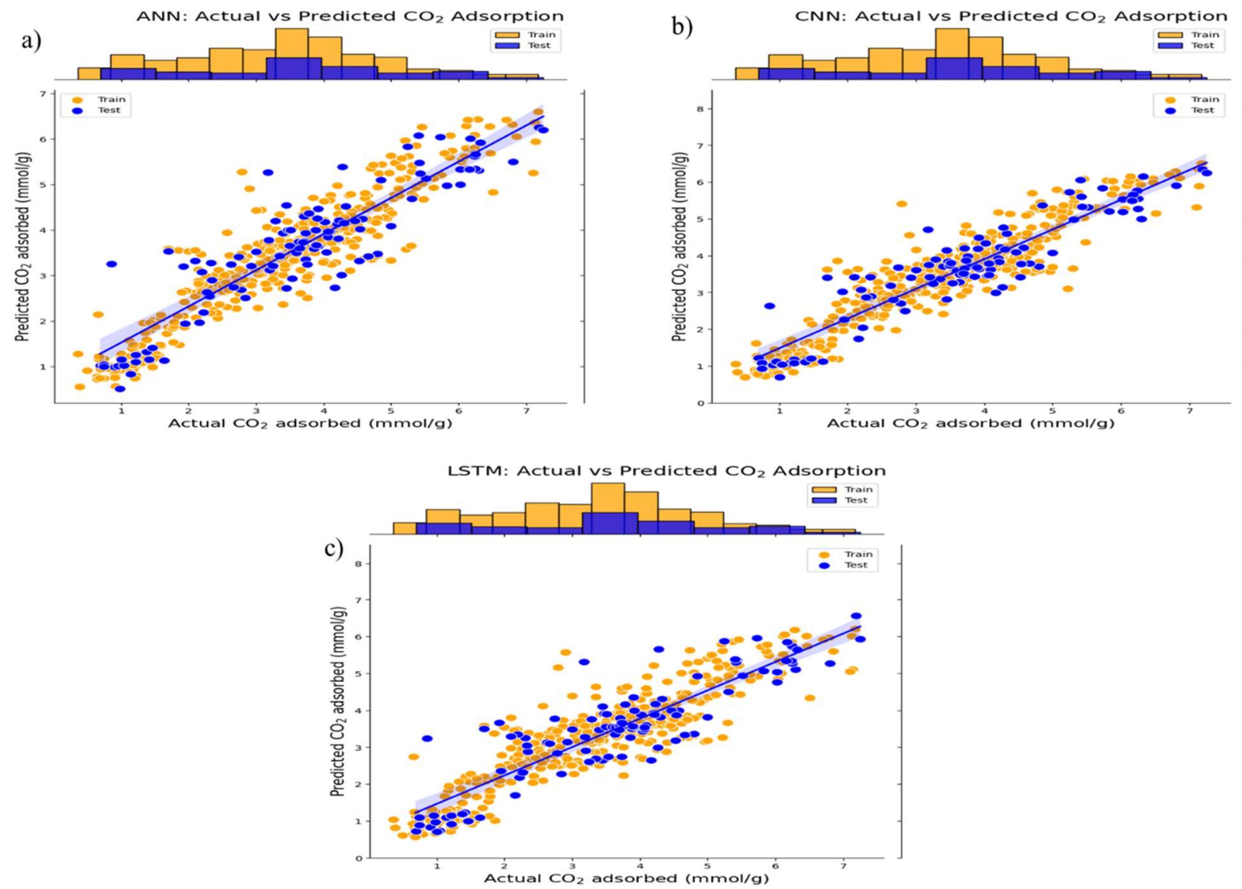


Table 3 presents a comparative overview of the predictive performance metrics for all five machine learning models. The Convolutional Neural Network (CNN) also delivered strong results, with training MSE = 0.2967, RMSE = 0.5448, MAE = 0.4123, R^2 = 0.9482, and validation R^2 = 0.9175, RMSE = 0.7565, reflecting its ability to model complex nonlinear relationships with minimal overfitting. The Long Short-Term Memory (LSTM) network showed moderate performance, with training MSE = 0.3492, RMSE = 0.5910, MAE = 0.4392, R^2 = 0.9380, and validation R^2 = 0.8995, RMSE = 0.8298, suggesting some sensitivity to data variability. The Artificial Neural Network (ANN) model, while capable of capturing dominant patterns during training (MSE = 0.3839, RMSE = 0.6196, MAE = 0.4541, R^2 = 0.9292), exhibited a larger drop in validation performance (MSE = 0.7718, RMSE = 0.8785, MAE = 0.6551, R^2 = 0.8919), indicating moderate overfitting and lower robustness. Overall, GBR and CNN were the most accurate and reliable models for predicting CO_2 adsorption capacity, followed by CNN, LSTM, and ANN, highlighting the effectiveness of ensemble tree methods and hybrid deep learning architectures in capturing complex interactions while maintaining strong generalization.

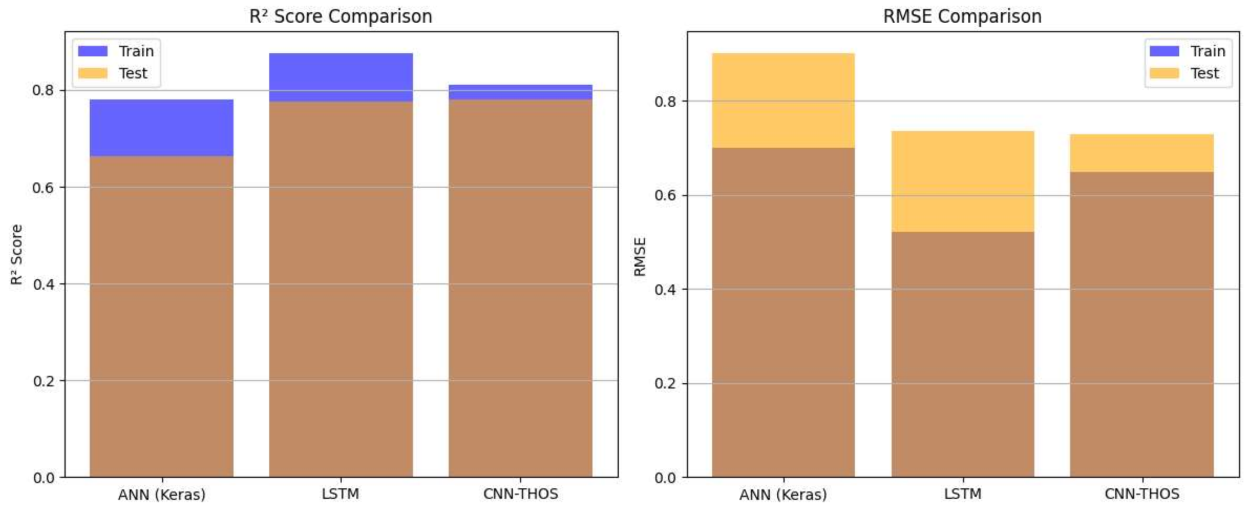


Figure . The comparison of the R^2, RMSE, CO_2 uptake for the four ML models.

Model Performance Comparison

MODEL	R^2 TRAIN	R^2 TEST	RMSE TRAIN	RMSE TEST
ANN	0.7793	0.6620	0.6992	0.9019
LSTM	0.8770	0.7758	0.5221	0.7347
CNN	0.8102	0.7799	0.6485	0.7278

3.4. Optimization Insights and Material Design

Leveraging the predictive power of the top-performing GBR model, 3D partial dependence plots were generated to visualize the combined effect of two features on CO_2 uptake while holding others constant. The analysis identified clear optimization pathways:

- **Surface Area:** An increase up to approximately 2000 m^2/g significantly enhances adsorption capacity.
- **Pore Size:** An optimal mean pore diameter range of 0–5 nm was identified, confirming the advantage of microporous structures for CO_2 capture at the studied conditions.

- **Pore Volume:** A pore volume above 0.5 cm^3/g was associated with higher uptake, facilitating better diffusion and access to adsorption sites.

These data-driven insights provide the actionable roadmap for synthesizing high-performance porous carbons, guiding the focus towards materials with high surface area, developed microporosity, and optimal pore volume, thereby reducing the need for extensive and costly trial-and-error experimentation.

Conclusion

In this study, a comprehensive machine learning workflow was successfully implemented to predict the CO_2 adsorption capacity of porous carbons. Among the models tested, the Convolutional Neural Network (CNN) and Gradient Boosting Regressor (GBR) delivered the most accurate and robust predictions, with the CNN achieving a superior test set R^2 of 85.73% ± 1.36. The analysis unequivocally identified operating conditions (temperature and pressure) and textural properties (surface area, pore volume, and pore size) as the primary determinants of adsorption performance. The creation of interaction-based features like the temperature-to-pressure ratio was instrumental in capturing the complex physics of the adsorption process. This work demonstrates that machine learning serves as a powerful tool to accelerate the discovery and design of next-generation adsorbents. By providing deep insights into the structure-property-performance relationships, ML models can guide experimental efforts, optimize synthesis parameters, and ultimately contribute to the development of more efficient and sustainable carbon capture technologies.

Reference

[1] A.M. Draper, M.J. Weissburg, Impacts of global warming and elevated CO2 on sensory behavior in predator-prey interactions: a review and synthesis, Front. Ecol.
Evol. 7 (2019) 72.

[2] F. Ahmad, Dynamic scheduling and control of MEA absorption processes for CO2 capture using gproms, Master’s thesis, University of Waterloo (2019).

[3] Ø.V. H. Lille-Mæhlum, Modelling solvent degradation in amine-based postcombustion carbon capture, Master’s thesis, NTNU (2021).

[4] T.C. Bond, S.J. Doherty, D.W. Fahey, P.M. Forster, T. Berntsen, B.J. DeAngelo, M. G. Flanner,

S. Ghan, B. K¨archer, D. Koch, et al., Bounding the role of black carbon in the climate system: a scientific assessment, J. Geophys. Res.: Atmos. 118 (11) (2013) 5380–5552.
[5] M.Z. Jacobson, Review of solutions to global warming, air pollution, and energy security, Energy Environ. Sci. 2 (2) (2009) 148–173.
[6] A. Raza, R. Gholami, R. Rezaee, V. Rasouli, M. Rabiei, Significant aspects of carbon capture and storage–a review, Petroleum 5 (4) (2019) 335–340.
[7] S. Zendehboudi, A. Khan, S. Carlisle, Y. Leonenko, Ex situ dissolution of CO2: a new engineering methodology based on mass-transfer perspective for enhancement of CO2 sequestration, Energy Fuels 25 (7) (2011) 3323–3333.
[8] Y. Yan, T.N. Borhani, S.G. Subraveti, K.N. Pai, V. Prasad, A. Rajendran, P. Nkulikiyinka, J.O. Asibor, Z. Zhang, D. Shao, et al., Harnessing the power of machine learning for carbon capture, utilisation, and storage (CCUS)–a state-ofthe-art review, Energy Environ. Sci. 14 (12) (2021) 6122–6157.

[9] M. Rahimi, M.H. Abbaspour-Fard, A. Rohani, O. YukselOrhan, X. Li, Modeling and optimizing N/O-enriched bio-derived adsorbents for CO2 capture: machine learning and DFT calculation approaches, Ind. Eng. Chem. Res. 61 (30) (2022)
10670–10688.
[10] K.M. Zorn, D.H. Foil, T.R. Lane, W. Hillwalker, D.J. Feifarek, F. Jones, W.D. Klaren, A.M. Brinkman, S. Ekins, Comparing machine learning models for aromatase (P450 19A1), Environ. Sci. Technol. 54 (23) (2020) 15546–15555.

[11] Breiman, L. Random Forests. *Machine Learning* **2001**, *45* (1), 5-32. DOI: 10.1023/A:1010933404324.

[12] Zhu, X.; Wang, X.; Ok, Y. S. The application of machine learning methods for prediction of metal sorption onto biochars. *Journal of Hazardous Materials* **2019**, *378*, 120727. DOI: 10.1016/j.jhazmat.2019.06.004.

[13] Probst, P.; Wright, M. N.; Boulesteix, A.-L. Hyperparameters and tuning strategies for random forest. *WIREs Data Mining and Knowledge Discovery* **2019**, *9* (3), e1301. DOI:
[14]Salehinejad, H., Sankar, S., Barfett, J., Colak, E., Valaee, S., 2017. Recent advances in recurrent neural networks. arXiv preprint arXiv:1801.01078.

[15]Shabbir, N., Kütt, L., Jawad, M., Amadiahanger, R., Iqbal, M.N., Rosin, A., 2019. Wind energy forecasting using recurrent neural networks. In: 2019 Big Data, Knowledge and Control Systems Engineering (BdKCSE), pp. 1–5.

[16] Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Comput. 9, 1735–1780.

[17]S. Malek, F. Melgani, Y. Bazi, One-dimensional convolutional neural networks for spectroscopic signal regression, J. Chemom. 32 (5) (2018), e2977.

[18] A. Ameri, M.A. Akhaee, E. Scheme, K. Englehart, Regression convolutional neural network for improved simultaneous EMG control, J. Neural Eng. 16 (3) (2019), 036015.

[19] R. Yamashita, M. Nishio, R.K.G. Do, K. Togashi, Convolutional neural networks: an overview and application in radiology, Insights Imaging 9 (2018) 611–629.

[20] J.L. Rodgers, W.A. Nicewander, Thirteen ways to look at the correlation coefficient, Am. Stat. (1988) 59–66.

[21]Yuan, X., Ge, Z., Song, Z., 2014. Soft sensor model development in multiphase/ multimode processes based on Gaussian mixture regression. Chemometr. Intell. Lab. Syst. 138, 97–109.